

Superstore data analysis

Basic

1. Find total sales and profit by region.
2. Find total sales and profit by category.
3. Find top 5 best-selling products by sales.
4. Find top 5 most profitable products.
5. Find average order value by customer segment.

Intermediate

6. Find sub-categories that are making a loss.
7. Find monthly sales trend.
8. Find top 10 states by sales.
9. Find average delivery days by ship mode.
10. Find high-discount orders that resulted in a loss.

Advanced

11. Get order, customer & product names via JOIN.
12. Find the top-selling product in each region.
13. Find running total of monthly sales.
14. Find customers who spend more than the average.
15. Find year-over-year sales growth %.

Query 1: Region-wise Total Sales aur Profit

```
select Region,  
sum(sales) as total_sales,  
sum(profit) as total_profit,  
count(*) as total_orders  
from Superstore_raw  
group by Region  
order by total_sales desc;
```

Results		Messages		
	Region	total_sales	total_profit	total_orders
1	West	725457.823114276	108418.789722558	3203
2	East	678781.237776518	91522.8399200849	2848
3	Central	501239.889593184	39706.4503734335	2323
4	South	391721.905365348	46749.7097326517	1620

Query 2: Category-wise Total Sales aur Profit

```

select category,
sum(sales) as total_sales,
sum(profit) as total_profit,
count(*) as total_orders
from Superstore_raw
group by category
order by total_sales desc;

```

	category	total_sales	total_profit	total_orders
1	Technology	836154.030862212	145455.660130084	1847
2	Furniture	741999.793703437	18451.2498190999	2121
3	Office Supplies	719047.031283677	122490.879799545	6026

Query 3: Top 5 Best-Selling Products (by Sales)

```

select top 5 [product_name],
sum(sales) as total_sales,
sum(quantity) as total_quantity
from Superstore_raw
group by [product_name]
order by total_sales desc;

```

	product_name	total_sales	total_quantity
1	Canon imageCLASS 2200 Advanced Copier	61599.822265625	20
2	Fellowes PB500 Electric Punch Plastic Comb Binding ...	27453.3840332031	31
3	Cisco TelePresence System EX90 Videoconferencing ...	22638.48046875	6
4	HON 5400 Series Task Chairs for Big and Tall	21870.5755004883	39
5	GBC DocuBind TL300 Electric Binding System	19823.478515625	37

Query 4: Top 5 Most Profitable Products

```

select top 5 [product_name],
sum(profit) as total_profit,
sum(sales) as total_sales
from Superstore_raw
group by [product_name]
order by total_profit desc;

```

	product_name	total_profit	total_sales
1	Canon imageCLASS 2200 Advanced Copier	25199.9406738281	61599.822265625
2	Fellowes PB500 Electric Punch Plastic Comb Bindin...	7753.05993652344	27453.3840332031
3	Hewlett Packard LaserJet 3310 Copier	6983.88989257813	18839.6859130859
4	Canon PC1060 Personal Laser Copier	4570.93994140625	11619.8341064453
5	HP DesignJet T520 Inkjet Large Format Printer - 24" ...	4094.97998046875	18374.8952636719

Query 5: Customer Segment-wise Average Order Value

```
select segment,
count(*) as total_orders,
avg(sales) as avg_order_value,
avg(sales) as tota_sales
from Superstore_raw
group by segment
order by avg_order_value desc;
```

	segment	total_orders	avg_order_value	tota_sales
1	Home Office	1783	240.972040763345	240.972040763345
2	Corporate	3020	233.823299163619	233.823299163619
3	Consumer	5191	223.733643555028	223.733643555028

Query 6: Loss-making Sub-Categories

```
select sub_category,
sum(profit) as total_profit,
sum(sales) as total_sales
from Superstore_raw
group by sub_category
having sum(profit) < 0
order by total_profit asc;
```

	sub_category	total_profit	total_sales
1	Tables	-17725.5901668072	206965.531606674
2	Bookcases	-3472.55999648571	114879.996025085
3	Supplies	-1188.98995436728	46673.5370957851

Query 7: Monthly Sales Trend

```
select year(order_date) as order_year,
month(order_date) as order_month,
sum(sales) as total_sales,
sum(profit) as total_profit
from Superstore_raw
group by year(order_date), month(order_date)
order by order_year, order_month;
```

Results		Messages		
	order_year	order_month	total_sales	total_profit
1	2014	1	14236.8949842453	2450.17996877432
2	2014	2	4519.89195883274	862.300012350082
3	2014	3	55691.0094711781	498.720014840364
4	2014	4	28295.344894886	3488.86003303528
5	2014	5	23648.2868702412	2738.74000123143
6	2014	6	34595.1274838448	4976.5599386692
7	2014	7	33946.3927909136	-841.459878444672
8	2014	8	27909.4681773186	5318.10996815562
9	2014	9	81777.3513105512	8328.07999272645
10	2014	10	31453.3927481174	3448.22994401678
11	2014	11	78628.7165292501	9292.15004178882
12	2014	12	69545.6202806234	8983.59004674107
13	2015	1	18174.0755178928	-3280.97008162737
14	2015	2	11951.4109165668	2813.84001851082
15	2015	3	38726.25218153	9732.06989204884
16	2015	4	34195.2084971666	4187.52995367348
17	2015	5	20121.6868264675	4667.82006281091

Query 8: State-wise Top 10 Sales

```
select top 10 state,
sum(sales) as total_sales,
sum(profit) as total_profit
from Superstore_raw
group by state
order by total_sales desc;
```

Results		Messages	
	state	total_sales	total_profit
1	California	457687.630218506	76381.5998399071
2	New York	310876.270583153	74038.6398655437
3	Texas	170188.045670152	-25729.2899689823
4	Washington	138641.269838929	33402.6998684406
5	Pennsylvania	116511.912908077	-15560.0400079042
6	Florida	89473.7084289789	-3399.24999237061
7	Illinois	80166.1008920074	-12607.8899646997
8	Ohio	78258.1357239485	-16971.3700857311
9	Michigan	76269.6139240265	24463.1500958353
10	Virginia	70636.7199046612	18598.0000146031

Query 9: Ship Mode-wise Average Delivery Days

```
select ship_mode,
avg(datediff(day,order_date,ship_date)) as avg_delivery_days,
count(*) as total_orders
from Superstore_raw
group by ship_mode
order by avg_delivery_days;
```

Results		Messages	
	ship_mode	avg_delivery_days	total_orders
1	Same Day	0	543
2	First Class	2	1538
3	Second Class	3	1945
4	Standard Class	5	5968

Query 10: High-Discount Orders with Negative Profit

```

select product_name,
discount,
sales,
profit
from Superstore_raw
where discount > 0.3 and profit < 0
order by discount desc;

```

	product_name	discount	sales	profit
1	Holmes Replacement Filter for HEPA Air Cleaner, Very ...	0.800000011920929	68.8099975585938	-123.860000610352
2	Storex DuraTech Recycled Plastic Frosted Binders	0.800000011920929	2.54399991035461	-3.8199999332428
3	Avery Hidden Tab Dividers for Binding Systems	0.800000011920929	1.78799998760223	-3.03999996185303
4	Economy Binders	0.800000011920929	1.24800002574921	-1.92999994754791
5	Kensington 7 Outlet MasterPiece Power Center	0.800000011920929	177.979995727539	-453.850006103516
6	Kensington 7 Outlet MasterPiece HOMEOFFICE Power ...	0.800000011920929	52.4480018615723	-131.119995117188
7	Acco 7-Outlet Masterpiece Power Center, Wihtout Fax/P...	0.800000011920929	97.2639999389648	-243.160003662109
8	Eureka Sanitaire Commercial Upright	0.800000011920929	66.2839965820313	-178.970001220703
9	Eureka Disposable Bags for Sanitaire Vibra Groomer I ...	0.800000011920929	1.62399995326996	-4.46999979019165
10	Round Ring Binders	0.800000011920929	2.07999992370605	-3.430000667572
11	Holmes Odor Grabber	0.800000011920929	8.65200042724609	-20.3299999237061
12	GBC Twin Loop Wire Binding Elements, 9/16" Spine, BL...	0.800000011920929	12.1759996414185	-18.8700008392334
13	Wilson Jones Century Plastic Molded Ring Binders	0.800000011920929	12.4619998931885	-20.5599994659424

Query 11: Get top 10 order, customer & product names via JOIN.

```

select top 10 o.Order_id, c.Customer_Name, p.Product_Name,
o.Sales, o.Profit, o.Region
from superstore_raw o
join Customers c on o.Customer_id = c.Customer_id
join Products p on o.Product_id = p.Product_id;

```

	Order_id	Customer_Name	Product_Name	Sales	Profit	Region
1	CA-2016-152156	Claire Gute	Bush Somerset Collection Bookcase	261.959991455078	41.9099998474121	South
2	CA-2016-152156	Claire Gute	Bush Somerset Collection Bookcase	261.959991455078	41.9099998474121	South
3	CA-2016-152156	Claire Gute	Bush Somerset Collection Bookcase	261.959991455078	41.9099998474121	South
4	CA-2016-152156	Claire Gute	Hon Deluxe Fabric Upholstered Stacking Chairs, R...	731.940002441406	219.580001831055	South
5	CA-2016-152156	Claire Gute	Hon Deluxe Fabric Upholstered Stacking Chairs, R...	731.940002441406	219.580001831055	South
6	CA-2016-152156	Claire Gute	Hon Deluxe Fabric Upholstered Stacking Chairs, R...	731.940002441406	219.580001831055	South
7	CA-2016-138688	Darrin Van Huff	Self-Adhesive Address Labels for Typewriters by Uni...	14.6199998855591	6.86999988555908	West
8	CA-2016-138688	Darrin Van Huff	Self-Adhesive Address Labels for Typewriters by Uni...	14.6199998855591	6.86999988555908	West
9	CA-2016-138688	Darrin Van Huff	Self-Adhesive Address Labels for Typewriters by Uni...	14.6199998855591	6.86999988555908	West
10	CA-2016-138688	Darrin Van Huff	Self-Adhesive Address Labels for Typewriters by Uni...	14.6199998855591	6.86999988555908	West

Query 12: Region-wise Top-Selling Product (RANK Window Function)

```

select Region, Product_Name, Total_Sales, Sales_Rank
from (
    select Region,
           Product_Name,
           sum(Sales) AS Total_Sales,
           rank() over (partition by Region order by sum(Sales) desc) as Sales_Rank
    from superstore_raw
    group by Region, Product_Name
) ranked
where Sales_Rank = 1;

```

	Region	Product_Name	Total_Sales	Sales_Rank
1	Central	Canon imageCLASS 2200 Advanced Copier	17499.94921875	1
2	East	Canon imageCLASS 2200 Advanced Copier	30099.9130859375	1
3	South	Cisco TelePresence System EX90 Videoconferencing ...	22638.48046875	1
4	West	Canon imageCLASS 2200 Advanced Copier	13999.9599609375	1

Query 13: Running Total of Monthly Sales

```
SELECT YEAR(Order_Date) AS Order_Year,
       MONTH(Order_Date) AS Order_Month,
       SUM(Sales) AS Monthly_Sales,
       SUM(SUM(Sales)) OVER (ORDER BY YEAR(Order_Date), MONTH(Order_Date)) AS Running_Total
FROM superstore_raw
GROUP BY YEAR(Order_Date), MONTH(Order_Date)
ORDER BY Order_Year, Order_Month;
```

Query 14: Customers Who Spend More Than the Average (Subquery)

```
SELECT Customer_ID, Customer_Name, SUM(Sales) AS Total_Spend
FROM superstore_raw
GROUP BY Customer_ID, Customer_Name
HAVING SUM(Sales) > (
    SELECT AVG(CustomerTotal.TotalSales)
    FROM (
        SELECT Customer_ID, SUM(Sales) AS TotalSales
        FROM superstore_raw
        GROUP BY Customer_ID
    ) CustomerTotal
)
ORDER BY Total_Spend DESC;
```

	Customer_ID	Customer_Name	Total_Spend
1	SM-20320	Sean Miller	25043.0504860878
2	TC-20980	Tamara Chand	19052.217195034
3	RB-19360	Raymond Buch	15117.3389587402
4	TA-21385	Tom Ashbrook	14595.6198644638
5	AB-10105	Adrian Barton	14473.571243763
6	KL-16645	Ken Lonsdale	14175.2289050817
7	SC-20095	Sanjit Chand	14142.3341379166
8	HL-15040	Hunter Lopez	12873.297662735
9	SE-20110	Sanjit Engle	12209.4381403923
10	CC-12370	Christopher Conant	12129.0715373755
11	TS-21370	Todd Sumrall	11891.7508912086
12	GT-14710	Greg Tran	11820.1200528145
13	BM-11140	Becky Martin	11789.6302245855

Query 15: Year-over-Year Sales Growth % (LAG Window Function)

```
SELECT Order_Year, Yearly_Sales,
       LAG(Yearly_Sales) OVER (ORDER BY Order_Year) AS Prev_Year_Sales,
       ROUND(
           (Yearly_Sales - LAG(Yearly_Sales) OVER (ORDER BY Order_Year)) * 100.0
           / LAG(Yearly_Sales) OVER (ORDER BY Order_Year), 2
       ) AS YoY_Growth_Percent
FROM (
    SELECT YEAR(Order_Date) AS Order_Year,
           SUM(Sales) AS Yearly_Sales
    FROM superstore_raw
    GROUP BY YEAR(Order_Date)
) yearly
ORDER BY Order_Year;
```

	Order_Year	Yearly_Sales	Prev_Year_Sales	YoY_Growth_Percent
1	2014	484247.497500002	NULL	NULL
2	2015	470532.50930351	484247.497500002	-2.83
3	2016	609205.596219122	470532.50930351	29.47
4	2017	733215.252826691	609205.596219122	20.36

